

Robotics 2

Detection and isolation of robot actuation faults (and collisions)

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Fault diagnosis problems - 1



- in the diagnosis of faults possibly affecting a (nonlinear) dynamic system various problems can be formulated
- Fault Detection
  - recognize that the malfunctioning of the (controlled) system is due to the occurrence of a fault (or not proper behavior) affecting some physical or functional component of the system
- Fault Isolation
  - discriminate which particular fault f has occurred out of a (large) class of potential ones, by distinguishing it from any other fault and from the effects of disturbances possibly acting on the system
- Fault Identification
  - determine the time profile (and/or class type) of the isolated fault f
- Fault Accommodation
  - modify the control law so as to compensate for the effects of the detected and isolated fault (possibly also identified)

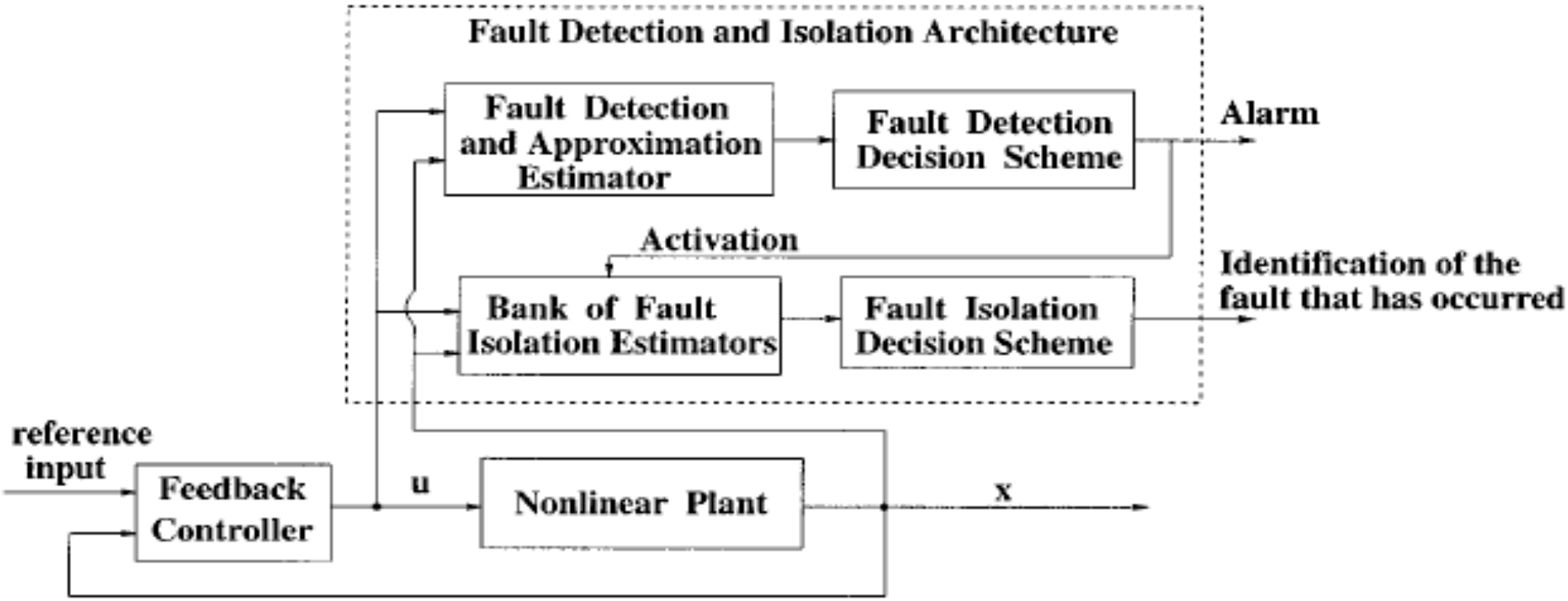
Fault diagnosis problems - 2



- FDI solution (simultaneous detection and isolation)
  - definition of an auxiliary dynamic system (Residual Generator) whose output will depend only on the presence of the fault f to be detected and isolated (and not on any other fault or disturbance) and will converge asymptotically to zero when $f \equiv 0$ (stability)
  - in case of many potential faults, each component r_i of the vector r of residuals will depend on one and only one associated fault f_i (possibly reproducing approximately its time behavior)
  - many of the FDI schemes are model-based: they use a nominal (fault- and disturbance-free) dynamic model of the system
- Fault Tolerant Control
  - passive: control scheme that is intrinsically robust to uncertainties and/or faults (typically having only moderate/limited effects)
  - active: control scheme involving a reconfiguration after FDI (with guaranteed performance for the faulted system)

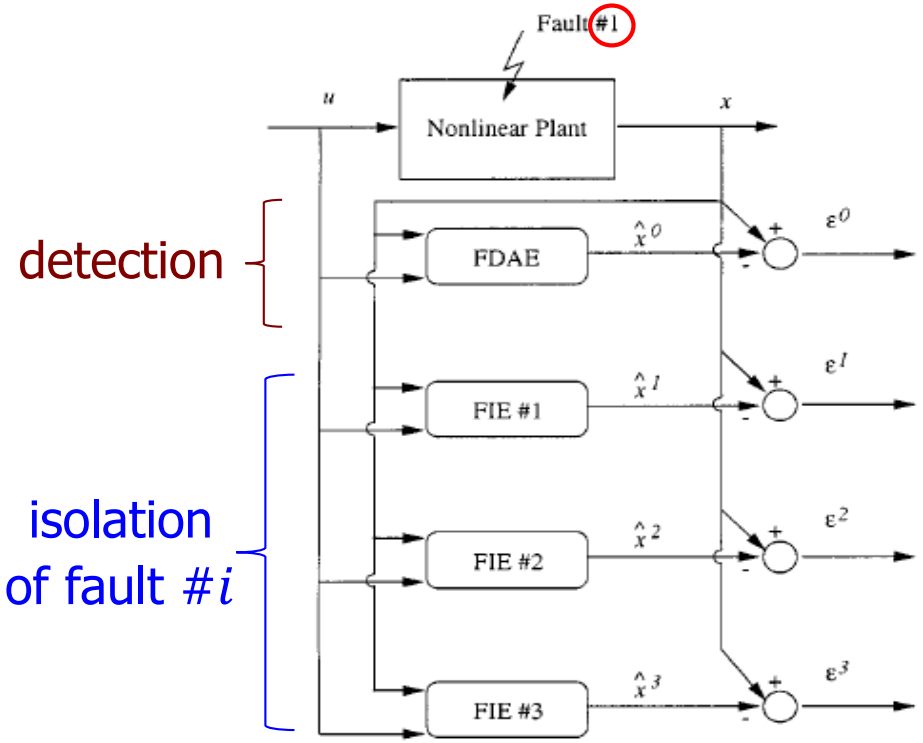
Typical FDI architecture

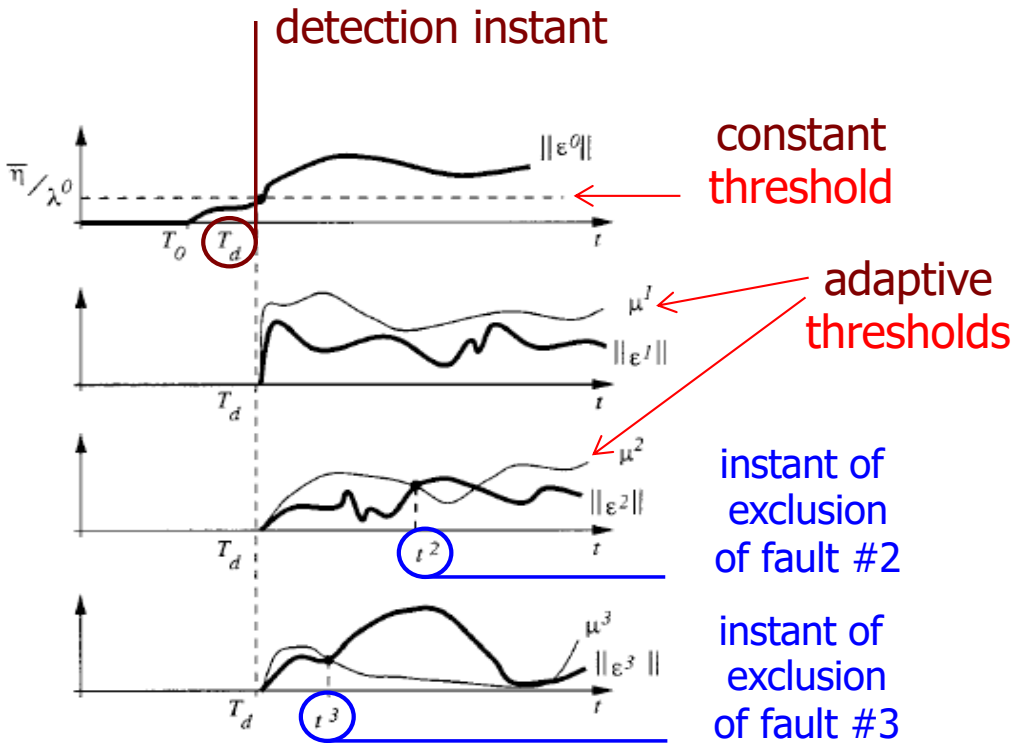
- bank of $n + 1$ (model-based) estimators
  - 1 for detection of a faulty condition
  - n for isolation of the specific (in general, modeled) fault



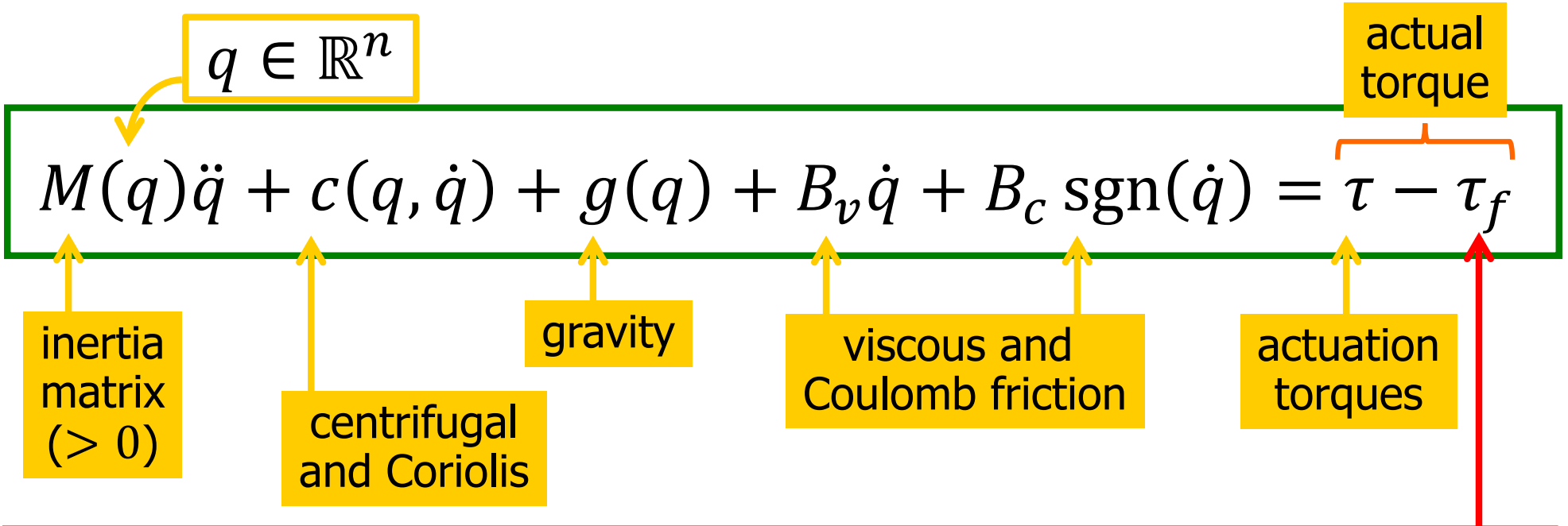
Some terminology

- fault types
  - instantaneous (abrupt), incipient (slow), intermittent, concurrent
- thresholds for detection/isolation (also adaptive)
  - delay times (from the instant T_0 of fault start) affect robot reaction
  - avoid false positives (useless alarms) and false negatives (missed events)





Actuator faults in robots



vector of actuation faults (even concurrent on more axes)

- total fault $\tau_{f,i} = \tau_i$
- partial fault $\tau_{f,i} = \varepsilon\tau_i \ (0 < \varepsilon < 1)$
- saturation $\tau_{f,i} = \tau_i - \mathrm{sgn}(\tau_i)\,\tau_{i,\max}$
- bias $\tau_{f,i} = b_i$
- block $\tau_{f,i} = \cdots$??
- ... any (collisions!)

Working assumptions



- signals and measurements available
  - the commanded input torque τ, but obviously not τ_f …
  - a measure of the full state $(q, \dot{q})$ is available
    - can be relaxed: in practice, with an estimate of joint velocities
  - no further sensors are anyway necessary ("sensorless")
- the robot dynamic model is known
  - in the absence of faults, and neglecting disturbances
  - no pre-specified model or type of faults is needed
- no dependence on/request of a specific input $\tau(t)$
  - can be anything (open loop, linear or nonlinear feedback)
- no dependence on/request of a specific motion $q_d(t)$

Generalized momentum

$$p = M(q)\dot{q}$$

with associated dynamic equation

$$\dot{p} = \tau - \tau_f - \alpha(q, \dot{q})$$

decoupled components relative to the single fault inputs

exploiting structure of centrifugal and Coriolis terms

$$\alpha_i = -\frac{1}{2}\dot{q}^T \frac{\partial M(q)}{\partial q_i}\dot{q} + g_i(q) + B_{vi}\dot{q}_i + B_{ci}\,\mathrm{sgn}(\dot{q}_i)$$

scalar expressions, for $i = 1, \cdots, n$

FDI solution

- definition of a vector of residuals

$$r = K\left[\int (\tau - \alpha(q,\dot{q}) - r)dt \; - \; p\right] \qquad K > 0 \text{ diagonal}$$

- no need to compute joint accelerations nor to invert the robot inertia matrix $M(q)$
- with perfect model knowledge, the dynamics of r is

$$\dot{r} = -Kr + K\tau_f$$

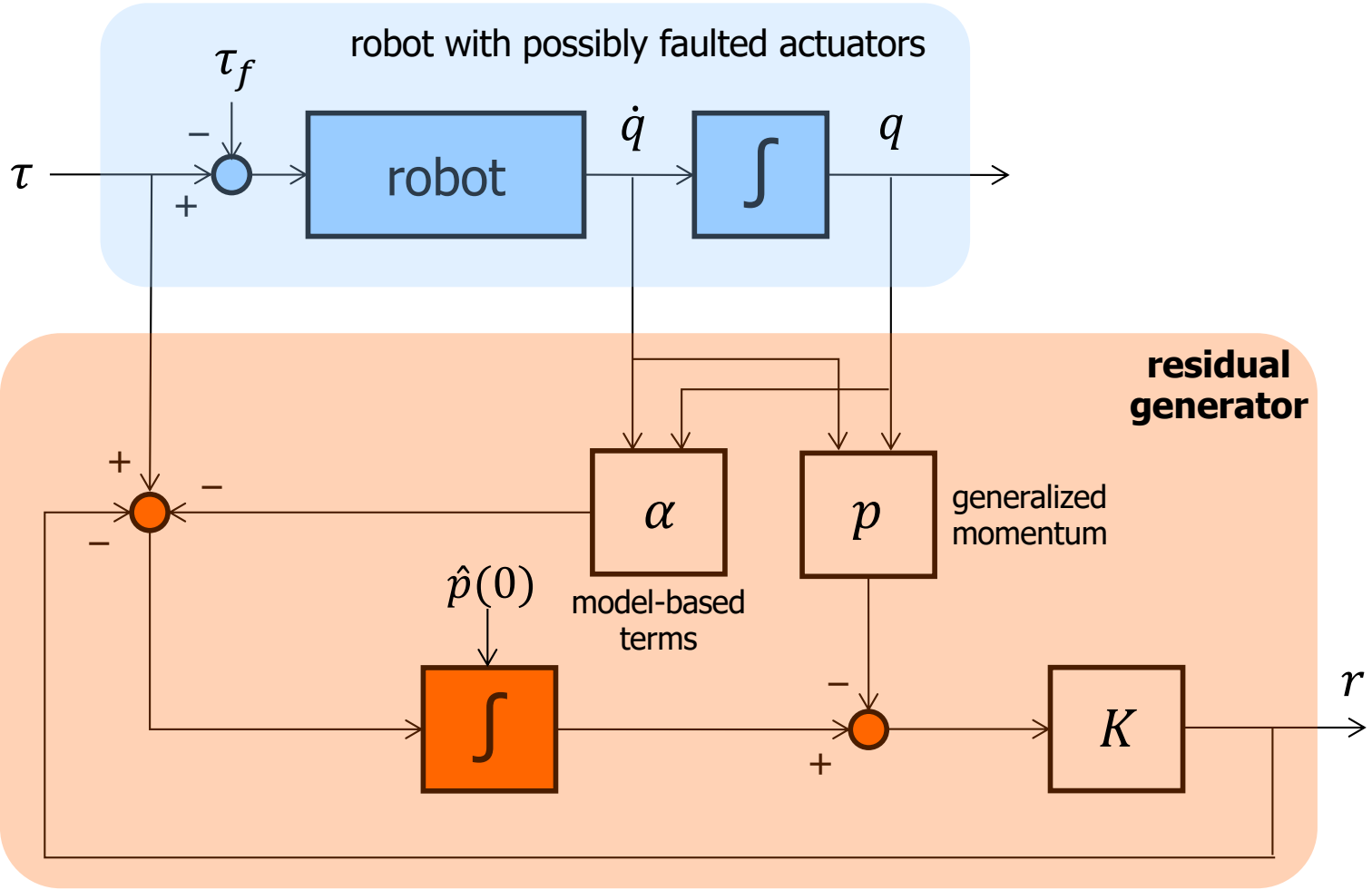
n decoupled filters, with unitary gains and time constants $1/k_i$

in the Laplace domain

$$\frac{r_i(s)}{\tau_{f,i}(s)} = \frac{k_i}{s + k_i} = \frac{1}{1 + \tau_i s}$$

for sufficiently large K, r reproduces the time behavior of τ_f

Block diagram of the residual generator



initialization of integrators $\hat{p}(0) = p(0)$ (zero if robot starts at rest)

$$r = K\left[\int (\tau - \alpha(q,\dot{q}) - r)dt \; - \; p\right]$$

Residual generator as "disturbance observer"



from the block diagram…

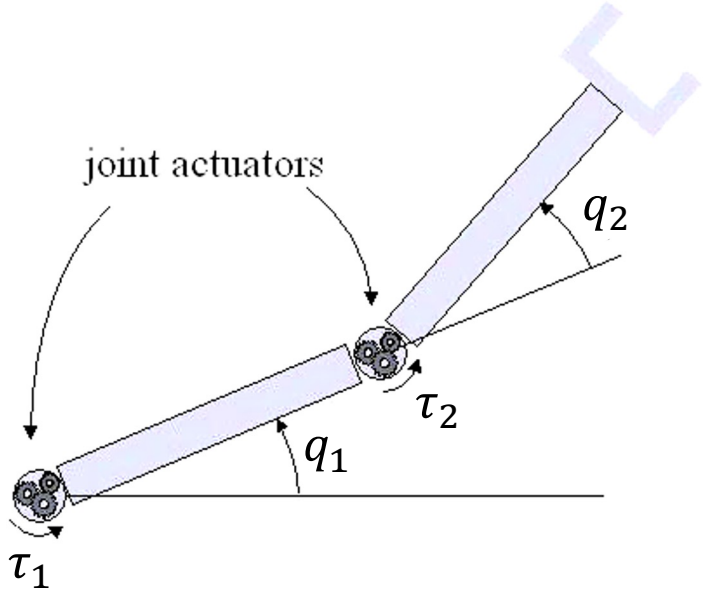
$$\dot{\hat{p}} = \tau - \alpha(q,\dot{q}) + K(p - \hat{p})$$
$$r = K(\hat{p} - p)$$

dynamic observer of the unknown actuation faults
($r \approx\!\!\rightarrow \tau_f$ = external disturbances)
with linear error dynamics (for constant τ_f)

$$e_{obs} = \tau_f - r \quad \Rightarrow \quad \dot{e}_{obs} = \dot{\tau}_f - \dot{r} = \dot{\tau}_f - K\left(\dot{\hat{p}} - \dot{p}\right)$$
$$= \dot{\tau}_f - K\left((\tau - \alpha - r) - (\tau - \alpha - \tau_f)\right)$$
$$= \dot{\tau}_f - K(\tau_f - r) = \dot{\tau}_f - K e_{obs} \cong -K e_{obs}$$

A worked-out example

- planar 2R robot under gravity



dynamic model (without friction)

$$M(q)\ddot{q} + \underbrace{c(q,\dot{q}) + g(q)}_{= C(q,\dot{q})\dot{q}} = \tau - \tau_f$$

$$\begin{pmatrix} a_1 + 2a_2c_2 & a_3 + a_2c_2 \\ a_3 + a_2c_2 & a_3 \end{pmatrix}\begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{pmatrix} + \begin{pmatrix} -a_2(2\dot{q}_1 + \dot{q}_2)\dot{q}_2 s_2 \\ a_2\dot{q}_1^2 s_2 \end{pmatrix} + \begin{pmatrix} a_4c_1 + a_5c_{12} \\ a_5c_{12} \end{pmatrix} = \begin{pmatrix} \tau_1 - \tau_{f,1} \\ \tau_2 - \tau_{f,2} \end{pmatrix}$$

computation of the residual vector

$$r = K\left[\int (\tau - \alpha(q,\dot{q}) - r)dt - p\right]$$

$$p = M(q)\dot{q}$$

$$\alpha_1 = g_1(q) = a_4c_1 + a_5c_{12}$$

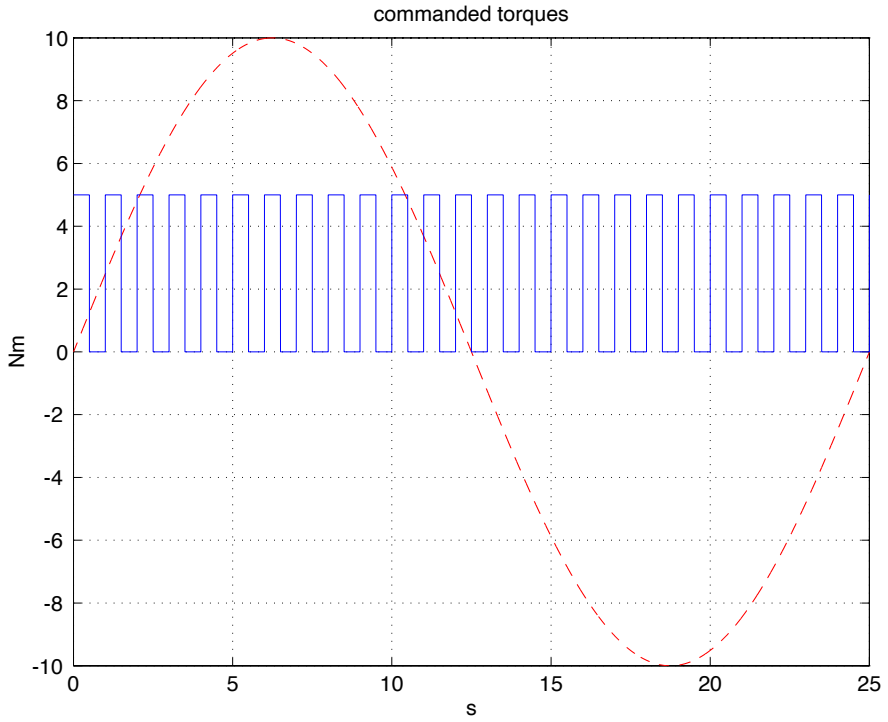
$$\alpha_2 = -\frac{1}{2}\dot{q}^T\frac{\partial M(q)}{\partial q_2}\dot{q} + g_2(q)$$

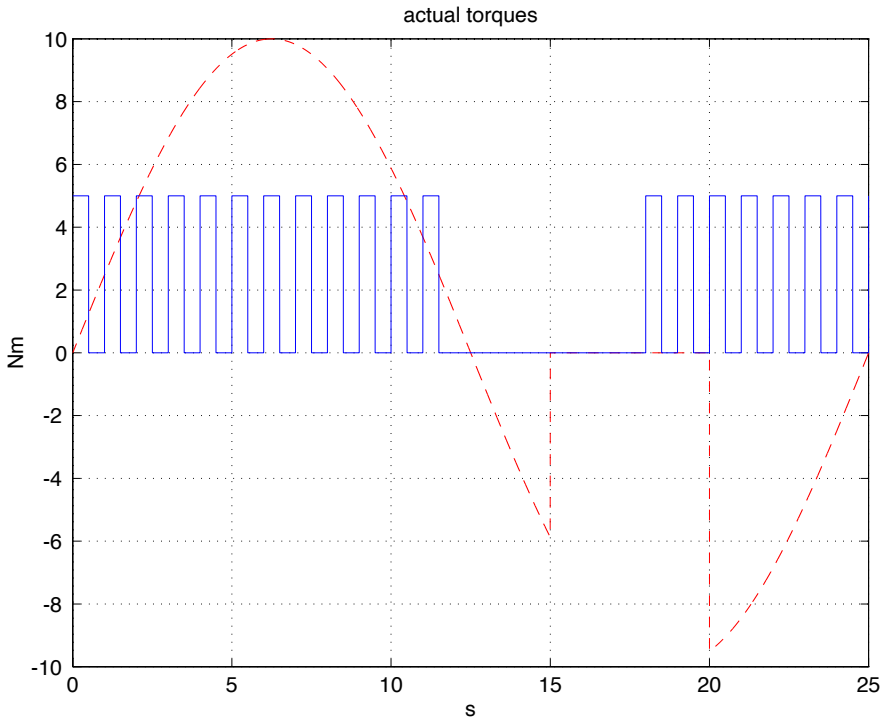
$$= a_2(\dot{q}_1 + \dot{q}_2)\dot{q}_1 s_2 + a_5c_{12}$$

Faults on both actuators
(total, intermittent, concurrent)

commanded torques (in open loop)

actual (faulted) torques





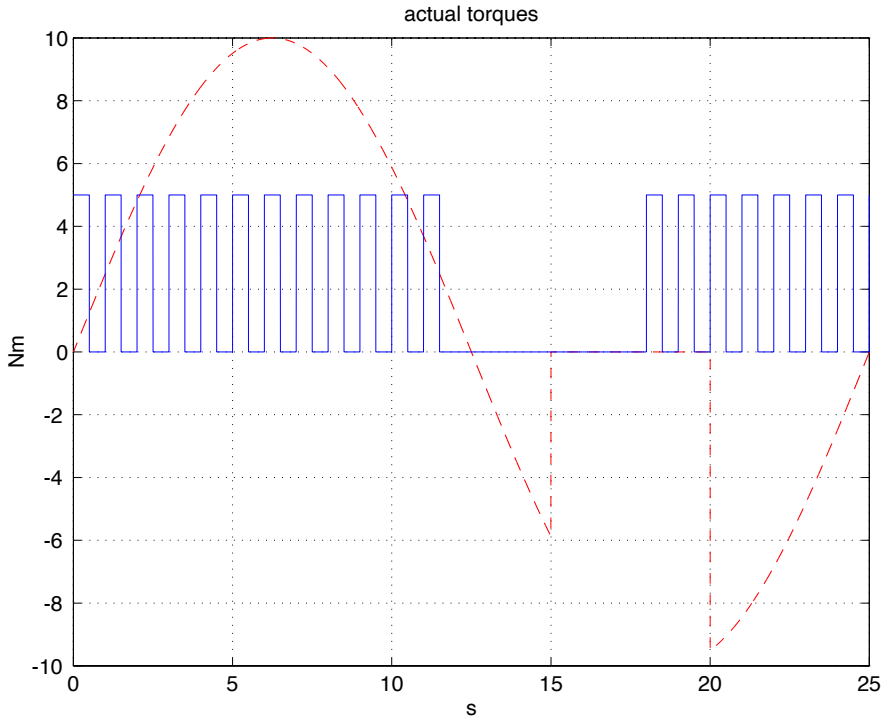
= first joint (fault for $t \in [15 \div 20]$ sec)

= second joint (fault for $t \in [12 \div 18]$ sec)

time interval of fault concurrence $t \in [15 \div 18]$ sec

First simulation

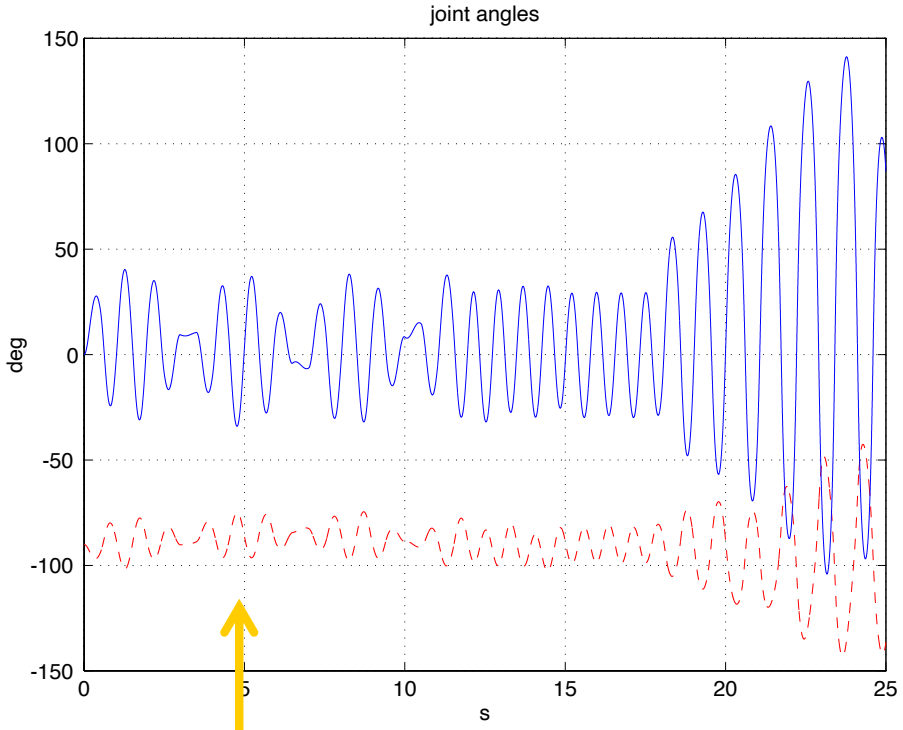
actual torques (to the robot)



= first joint

= second joint

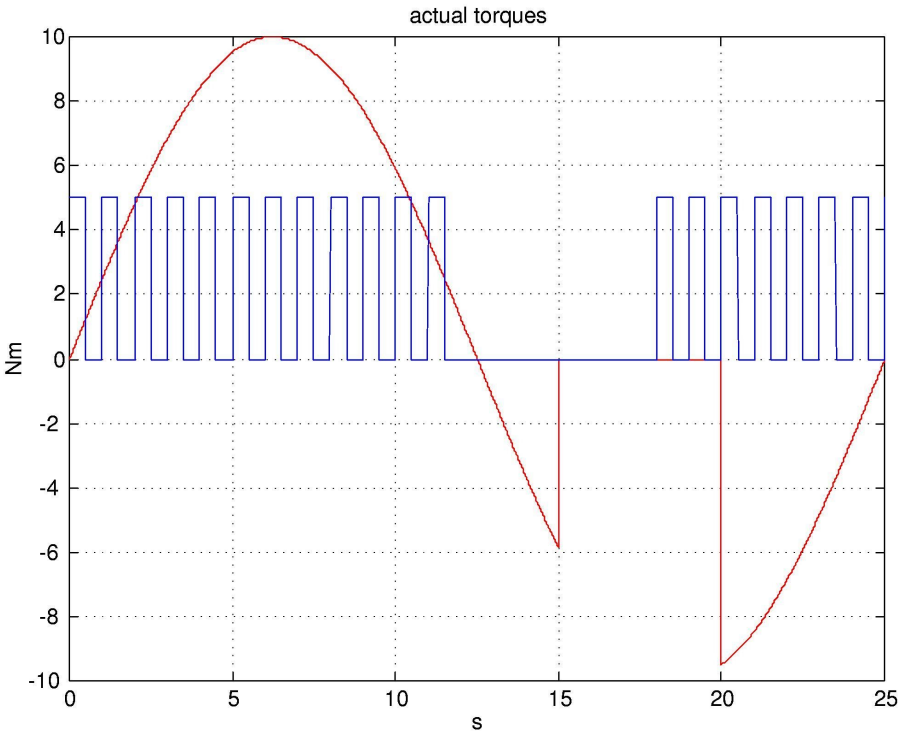
(measured) joint positions



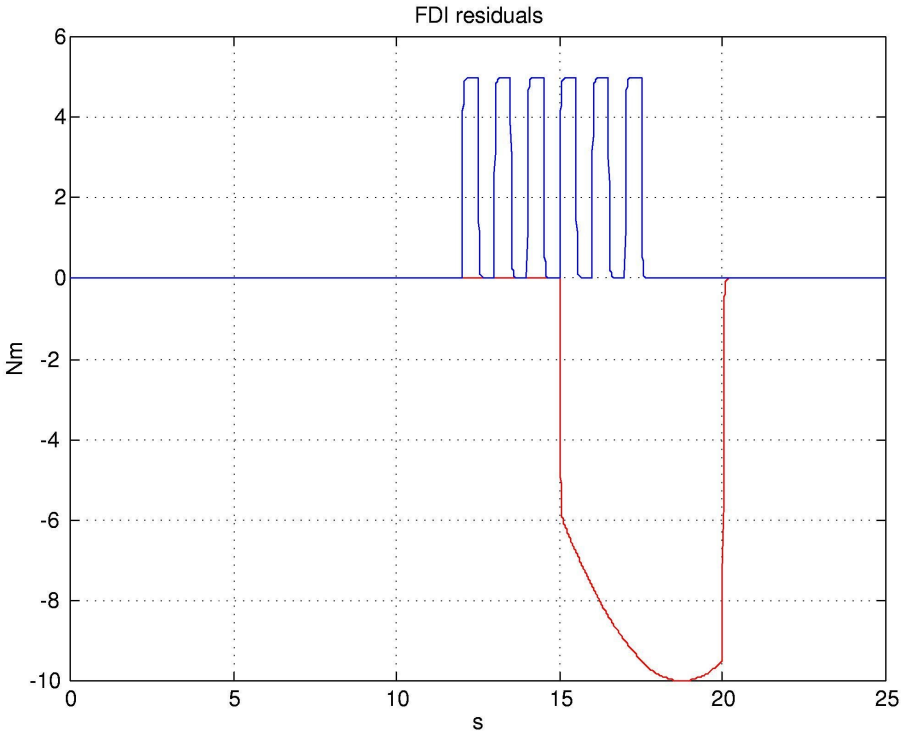
no clear evidence of faults in the dynamic evolution of the system!

First simulation – FDI

actual torques (to the robot)



residuals



= first joint

= second joint

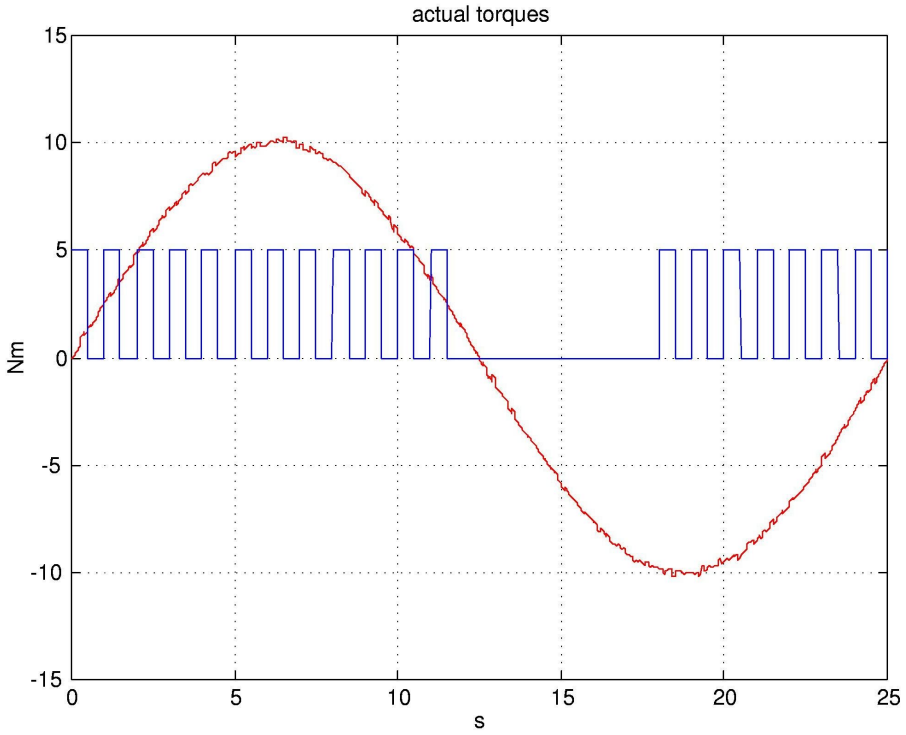
$$K = \mathrm{diag}\{50, 50\}$$

residuals reconstruct the "missing" parts of the torques (identification property!)

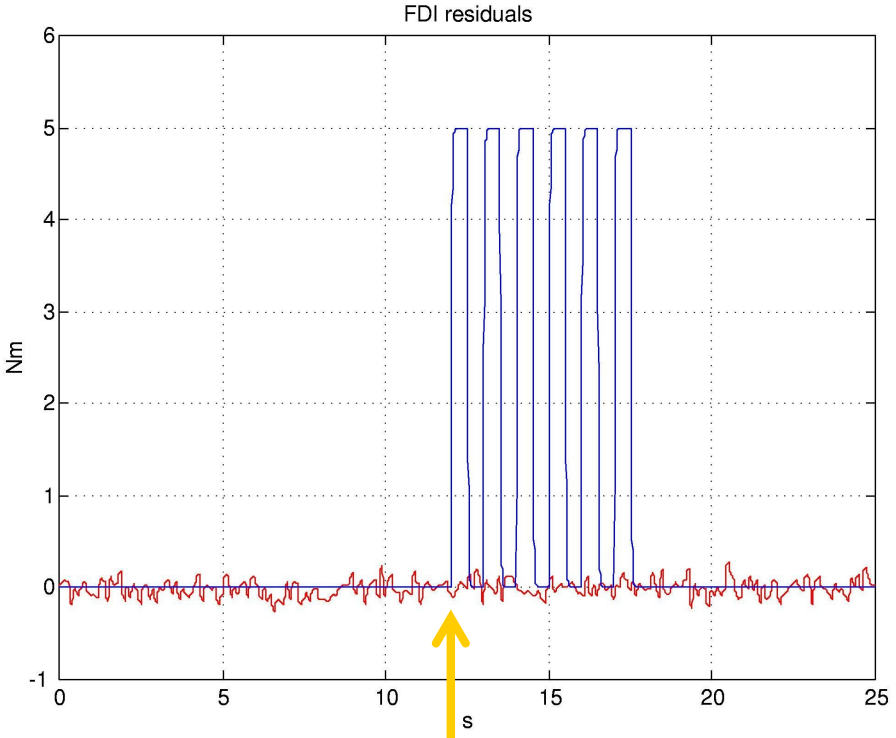
Second simulation – FDI

(total fault on second actuator, added noise on first channel)

actual torques (to the robot)



residuals



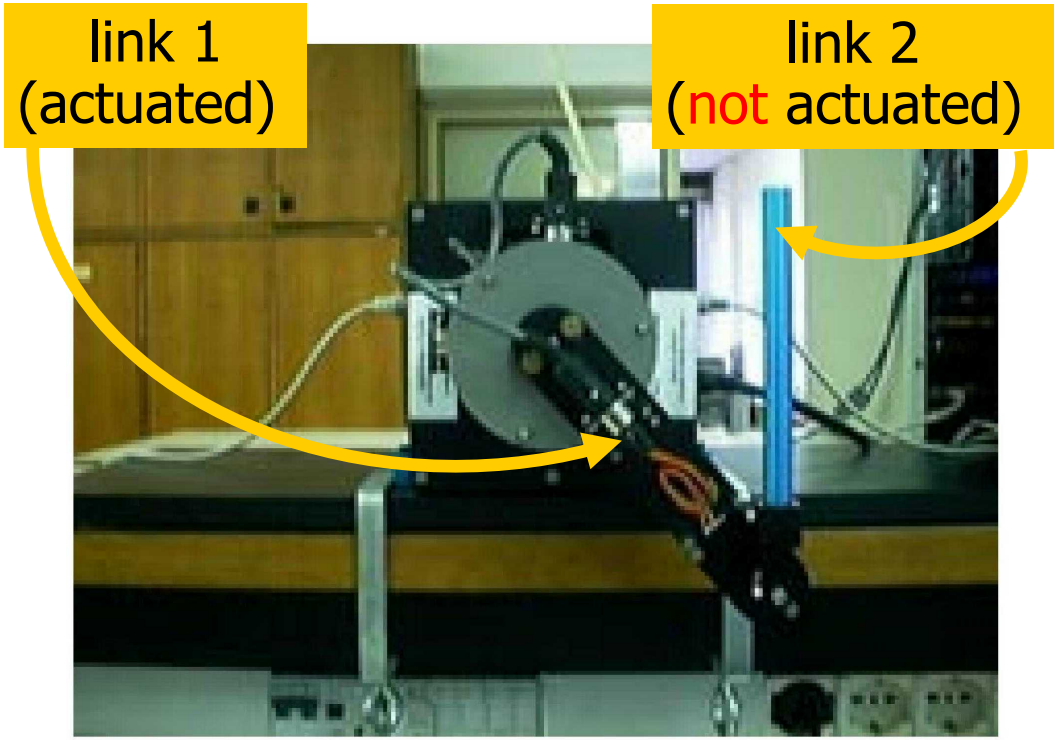
= first joint

= second joint (fault for $t \in [12 \div 18]$ sec)

residual r_1 is not affected by faulty actuation, while residual r_2 is not affected by the disturbance on first channel (decoupling property)

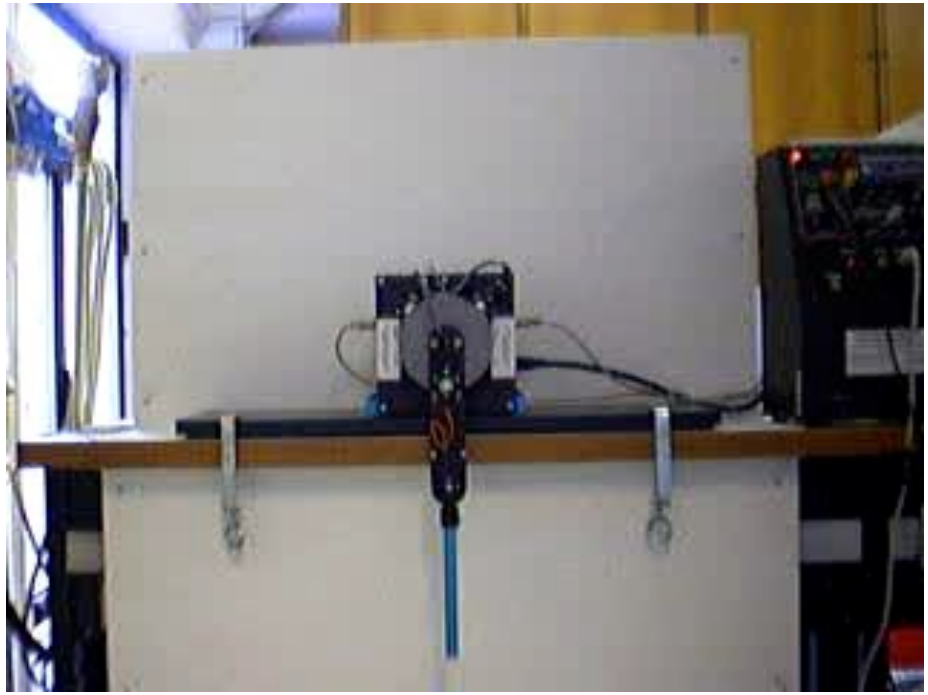
Experimental setup

Quanser Pendubot



with encoders on both joints

video



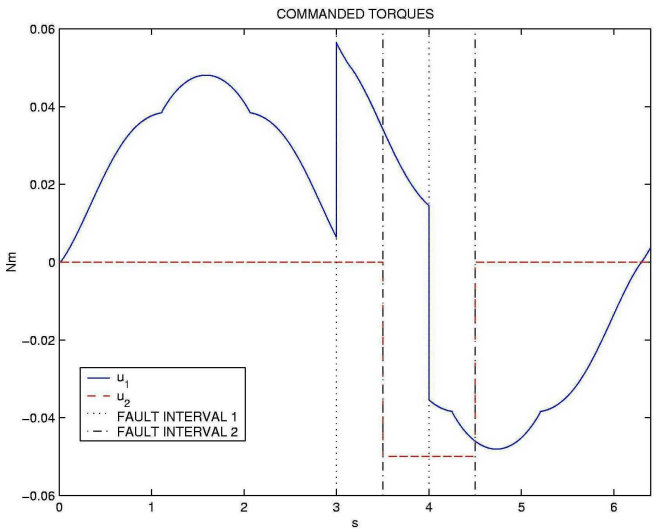
nonlinear control for swing-up

sampling time $T_c = 1$ ms, residual gains $K_i = 50$,
practical thresholds for fault detection $\cong 10^{-2} \div 10^{-3}$ Nm

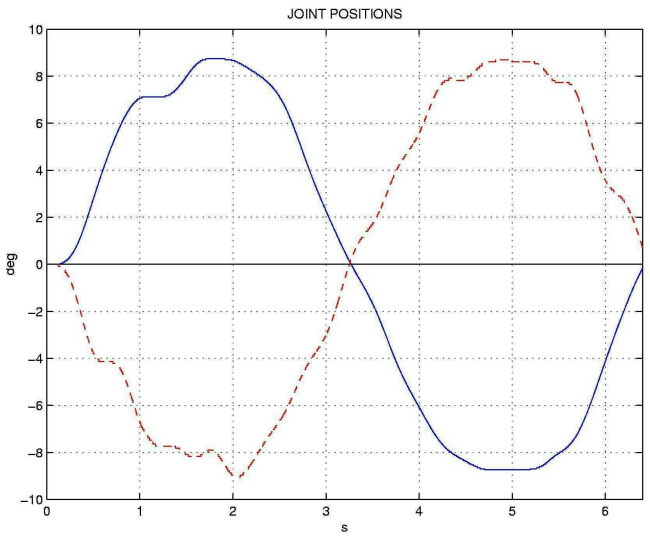
First experiment

- motor 1 driven by sinusoidal voltage of period 2π sec (open loop)
- bias fault on τ_1 for $t \in [3 \div 4]$ sec
- total fault on second joint for $t \in [3.5 \div 4.5]$ sec (a constant torque is requested, but no motor available at the joint to provide 0.05 Nm …)
- fault concurrency for $t \in [3.5 \div 4]$ sec

commanded torques



joint positions

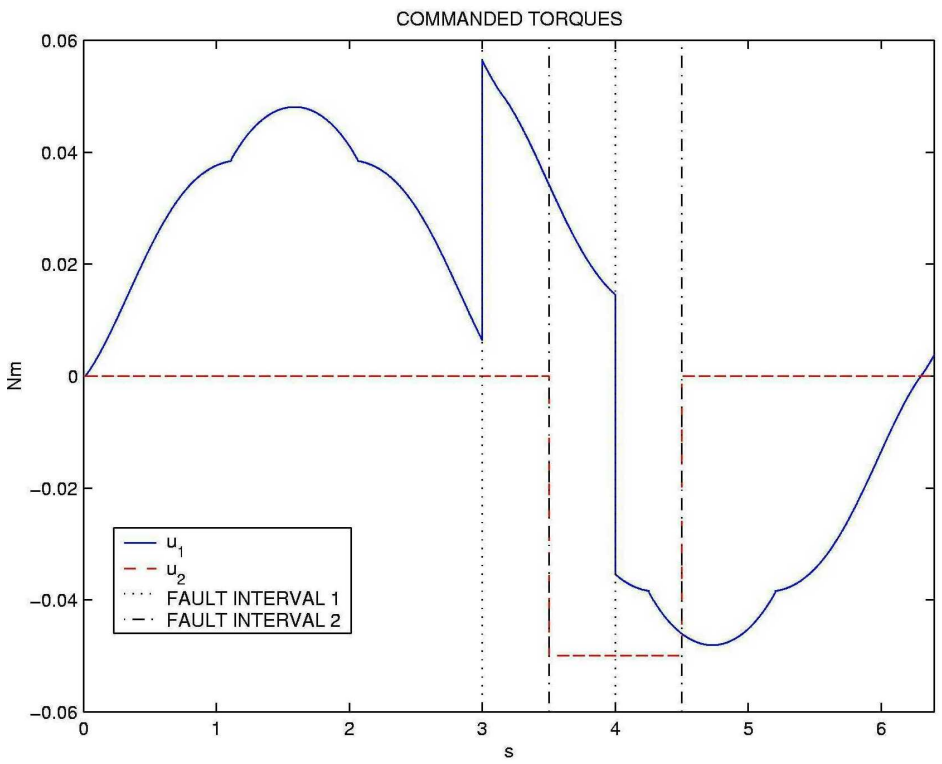


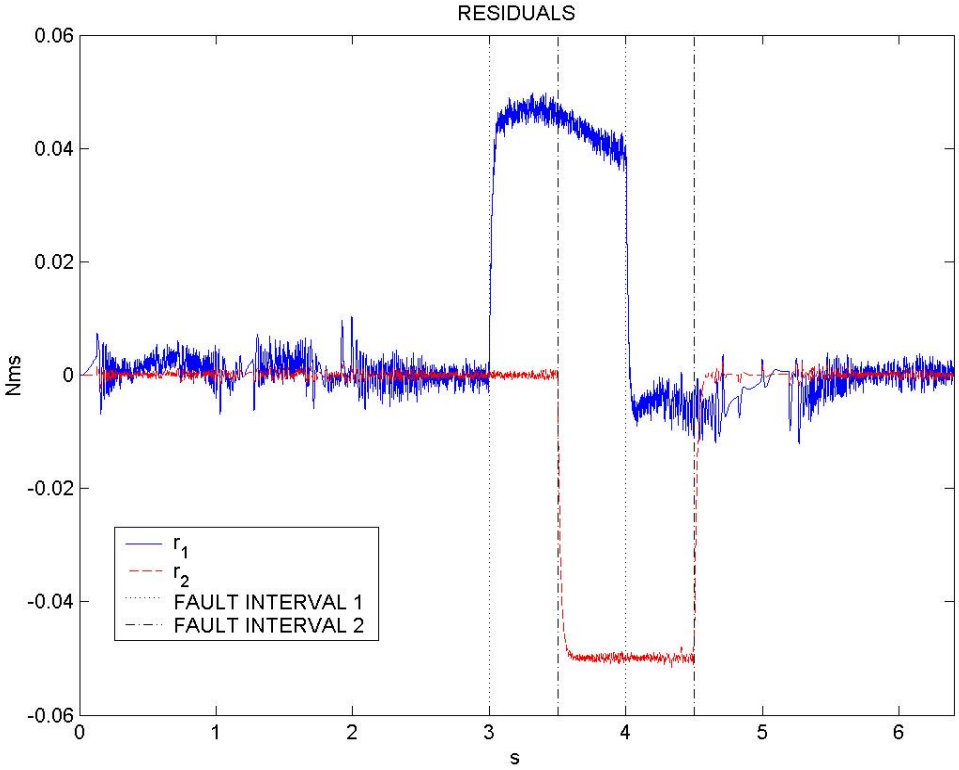
joint 1 joint 2

First experiment – FDI

commanded torques

residuals



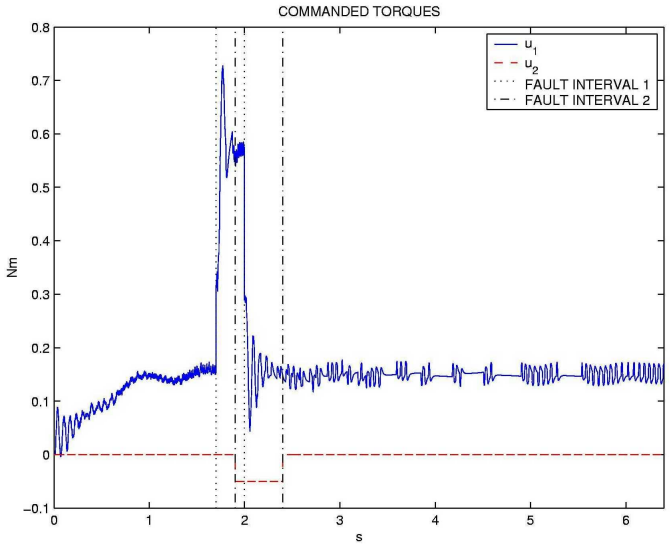


joint 1 joint 2

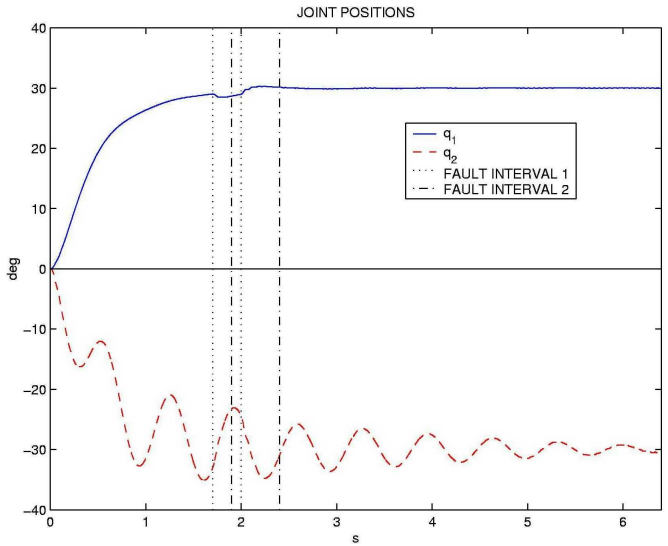
Second experiment

- position regulation of the first joint at $q_{d1} = 30°$ (under PID control)
- 50% power loss on motor 1 for $t \in [1.7 \div 2]$ sec
- total fault on joint 2 for $t \in [1.9 \div 2.4]$ sec (no motor …)
- fault concurrency for $t \in [1.7 \div 1.9]$ sec

commanded torques



joint positions

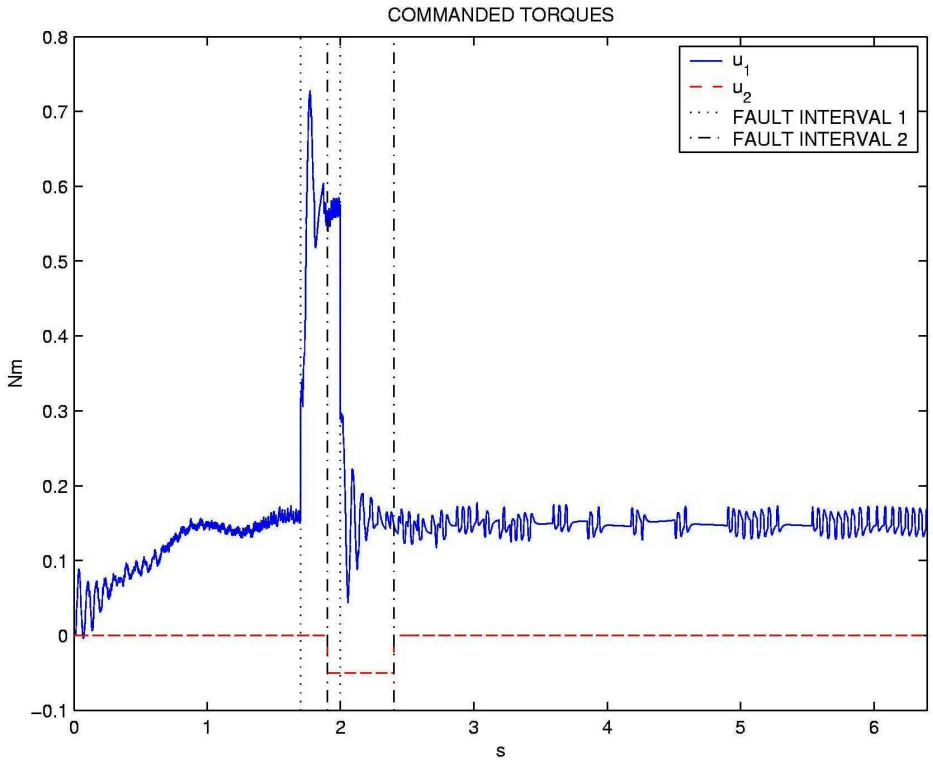


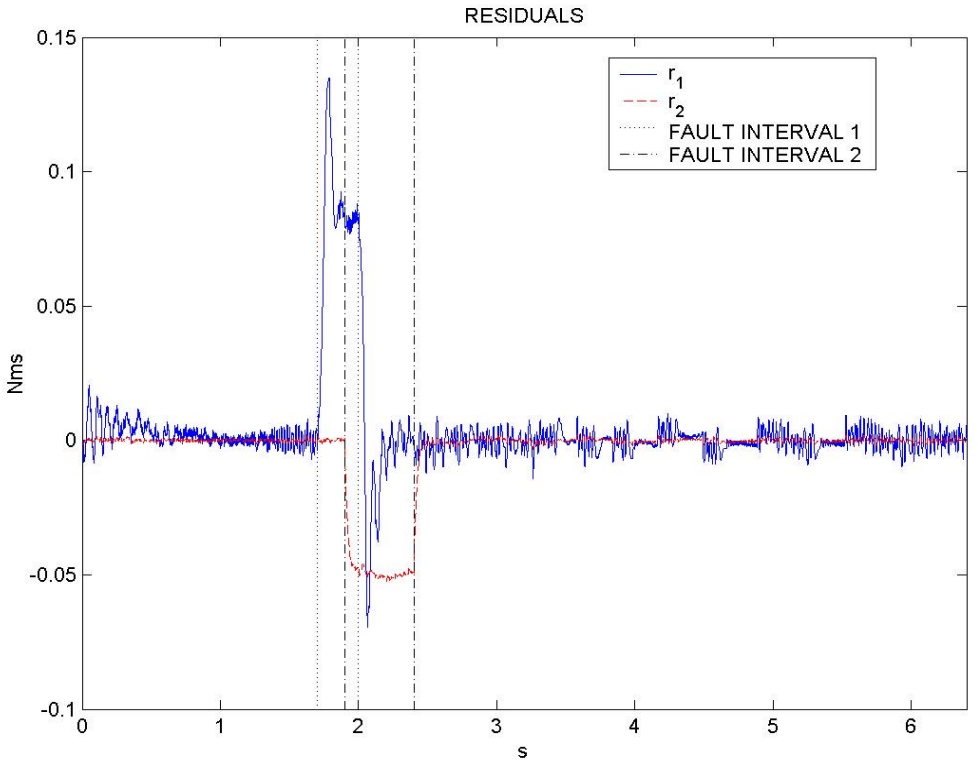
joint 1 joint 2

Second experiment – FDI

commanded torques

residuals



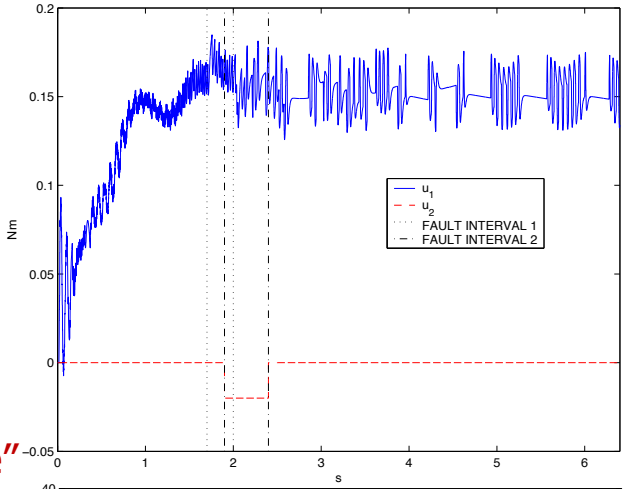


joint 1 joint 2

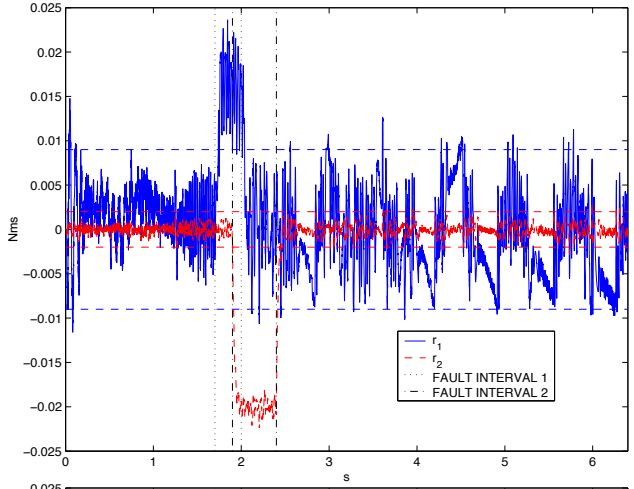
Third experiment – FDI

- same as in second experiment, but with only 10% power loss on motor 1
  - due to noisy PWM signals driving the DC motor, a dynamic filtering of residuals is used, staying above [below] a threshold ($r_{1,thres} = 9 \cdot 10^{-3}$ Nm, $r_{2,thres} = 2 \cdot 10^{-3}$ Nm) for a time $T_{set} = 0.02$ s [$T_{reset} = 0.03$ s] before detecting a fault [reset to normal operation]

commanded torques



residuals

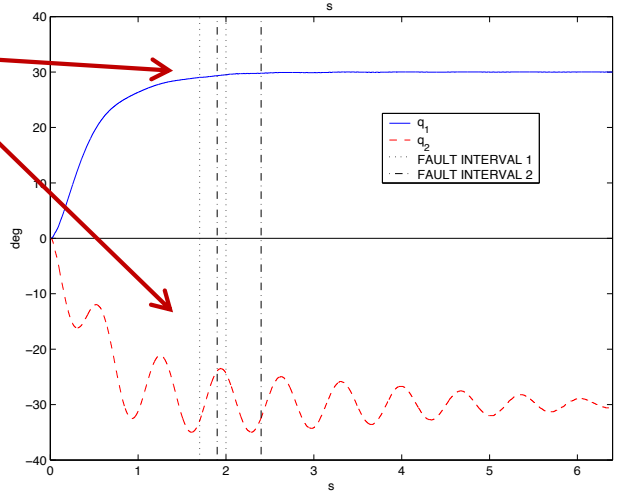


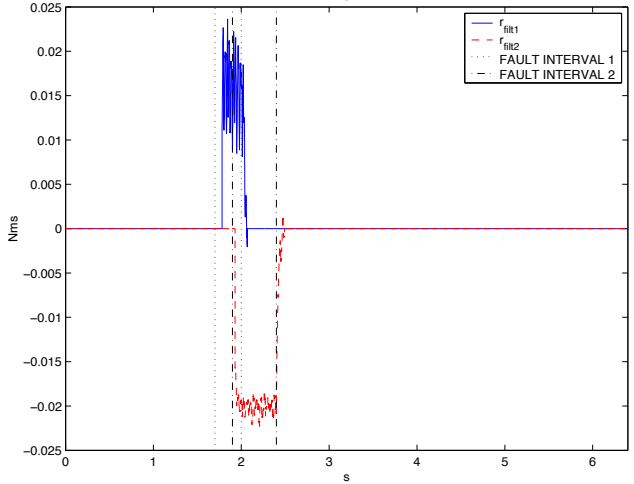
joint 1

joint 2

faults seem "unobservable" in these evolutions

joint positions





filtered residuals

Some extensions

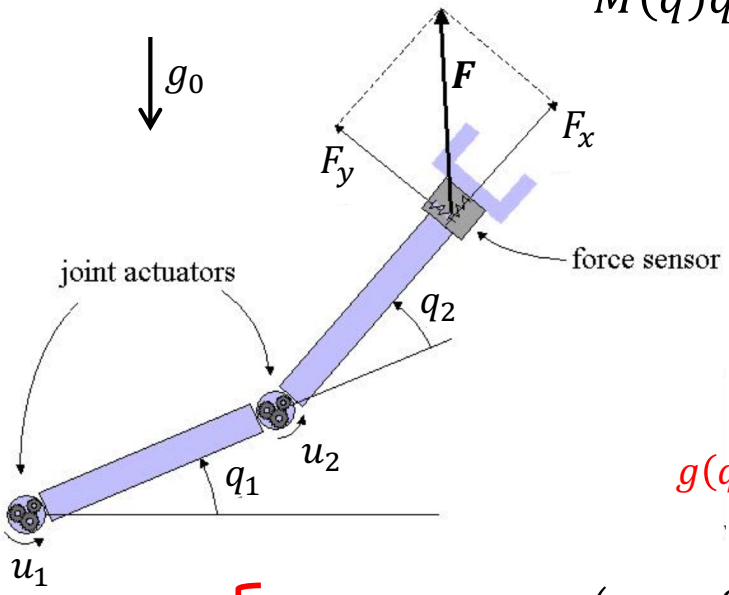


- FDI method based on generalized momentum is easily extended to the presence of flexible transmissions (elastic joints), actuator dynamics, ...
- the scheme can be made adaptive, in order to handle parametric uncertainties in the robot dynamic model
- the method can be modified for detection and isolation of significant classes of sensor faults (e.g., faults in force/torque sensor at the wrist)
  - applies to all faults that instantaneously affect robot acceleration or torque
- assuming non-concurrency (at most a single fault occurs at the same time) of a given set of faults, relaxed FDI conditions have been derived
  - of interest when the necessary conditions for multiple FDI are violated
  - involves processing of continuous residuals + discrete logic for isolation
- the same FDI-type approach has been applied also for compensation of unmodeled friction (treated as a "permanent fault" on the system)
- extended to robot collision detection and isolation (for safe HRI!)
- combination of model- and signal-based approaches to FDI

Isolation of F/T sensor faults

- planar 2R robot with fault on force measure of sensor on the end-effector

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau + J^T(q)F = \tau + J^T(q)(F_m + f_F)$$



$$J(q) = \begin{pmatrix} \ell_1 s_2 & 0 \\ \ell_2 + \ell_1 c_2 & \ell_2 \end{pmatrix}$$

robot Jacobian expressed in end-effector frame

$$J^{ad}(q) = \begin{pmatrix} \ell_2 & 0 \\ -\ell_2 - \ell_1 c_2 & \ell_1 s_2 \end{pmatrix}$$

adjoint of Jacobian
$J^{ad} = \det(J) \cdot J^{-1}$
⇒ a singularity robust scheme!

residual generator (function of $q, \dot{q}, F_m, \zeta$)

$$\dot{\zeta} = -(J^T)^{ad}(q)\begin{pmatrix} a_4 c_1 + a_5 c_{12} \\ a_5 c_{12} + a_2 s_2 \dot{q}_1(\dot{q}_1 + \dot{q}_2) \end{pmatrix} + \begin{pmatrix} 0 & \ell_1 s_2 \dot{q}_2 \\ 0 & \ell_1 c_2 \dot{q}_2 \end{pmatrix} M(q)\dot{q} + (J^T)^{ad}(q)\tau$$

$$+ \ell_1 \ell_2 s_2 F_m + K[\,(J^T)^{ad}(q)M(q)\dot{q} - \zeta]$$

$$r = (J^T)^{ad}(q)M(q)\dot{q} - \zeta$$

$g(q) - C^T(q,\dot{q})\dot{q}$ ↓

time derivative of transposed Jacobian adjoint ↓

robot inertia ↓

input torque ↓

$\det J^T(q)$ ↑

measured force (nominal) ↑

predicted FDI behavior in presence of force sensor faults $f_F \in \mathbb{R}^2$ ➡

$$\dot{r} = -Kr + \ell_1 \ell_2 \sin q_2 \, f_F$$

decoupled, though modulated by q_2

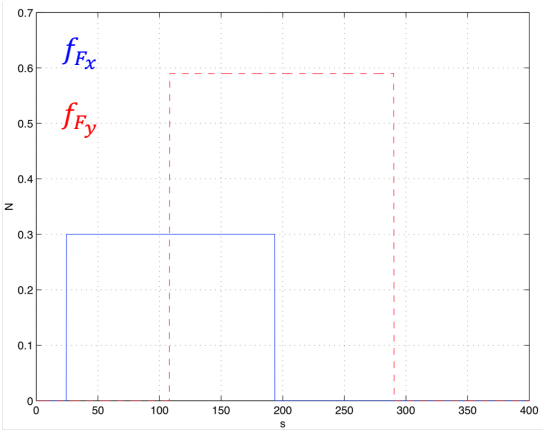
homework: prove this expression

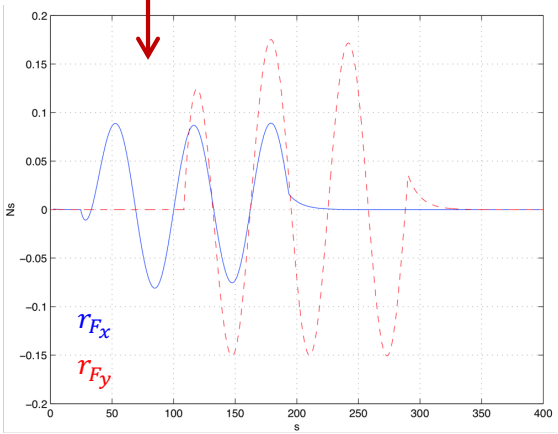
Isolation of F/T sensor faults

- simulation on the 2R robot

q_2 is tracking a sinusoid ($A = \pi/8$ rad, $\omega = 0.1$ rad/s)

bias faults on both components of force sensor
$f_{F_x} = 0.3$ N for $t \in [25 \div 190]$
$f_{F_y} = 0.6$ N for $t \in [109 \div 285]$

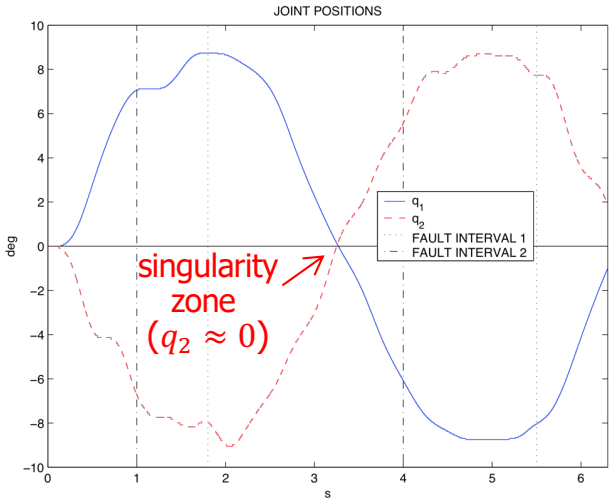


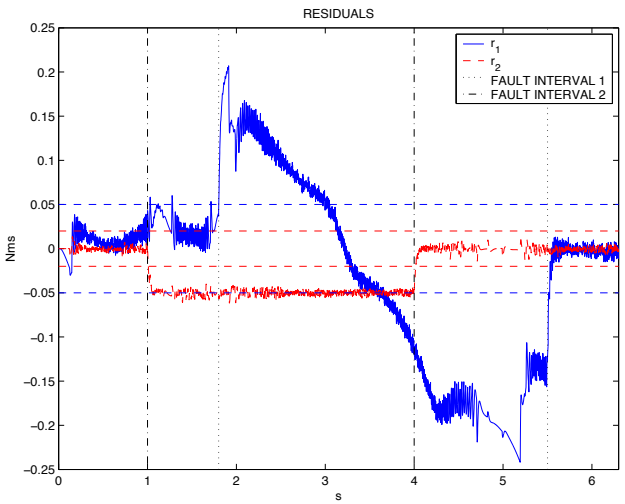


FDI residual components (with $K = 0.1I$)

- experiment on the Pendubot (no force sensor and no contact!)

evolution of joint variables





residuals response to emulated bias faults on force measures
-1 N on F_x in $t \in [1.8 \div 5.5]$
0.05 N on F_y in $t \in [1 \div 4]$

$(J^T)^{ad} \longrightarrow \mathrm{diag}\{s_2, 1\}\, J^{-T}$
in previous scheme

Isolation of non-concurrent faults

- faults of actuators AND faults of force sensor components (possibly occurring simultaneously) CANNOT be detected AND isolated
  - for a mechanical system with n dofs, the max # of faults allowing FDI is n!
- with non-concurrency, e.g., 2 actuator + 2 F/T sensor faults in 2R robot

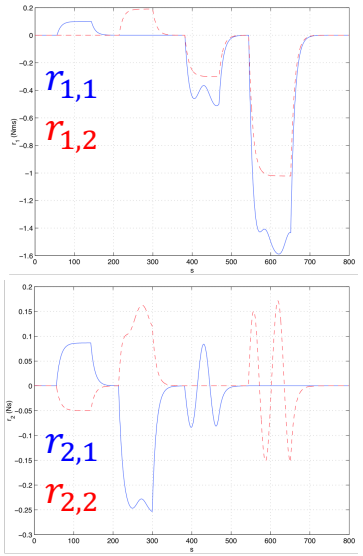
dependence of residuals on considered faults

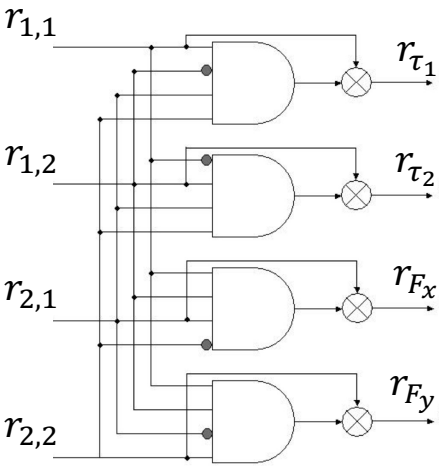
residual / fault	$r_{1,1}$	$r_{1,2}$	$r_{2,1}$	$r_{2,2}$
f_{τ_1}	1	0	1	1
f_{τ_2}	0	1	1	1
f_{F_x}	1	1	1	0
f_{F_y}	1	1	0	1

isolation matrix

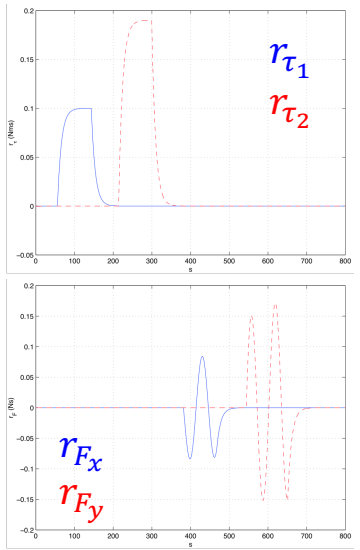
$r_{1,1}$ $r_{1,2}$ \ $r_{2,1}$ $r_{2,2}$	11	10	01	00
10	f_{τ_1}	NA	NA	NA
01	f_{τ_2}	NA	NA	NA
11	NC	f_{F_x}	f_{F_y}	NA
00	NA	NA	NA	no fault

time sequence of non-concurrent bias faults:
$f_{\tau_1} \rightarrow f_{\tau_2} \rightarrow f_{F_x} \rightarrow f_{F_y}$





isolation logics



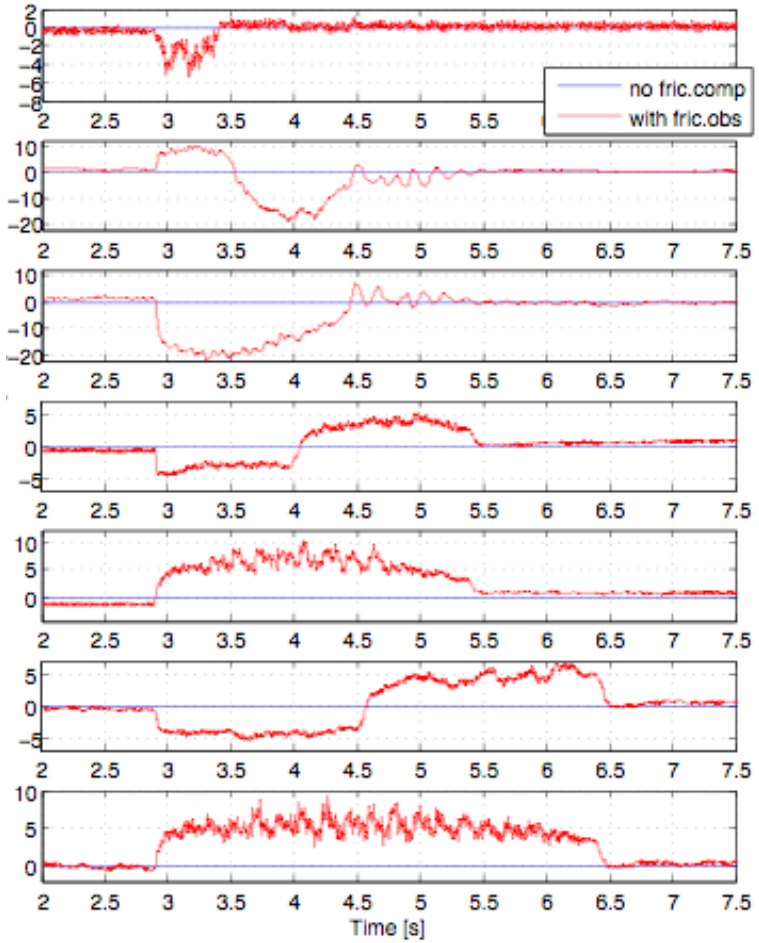
hybrid residuals (in continuous time + digital logics) allow **isolation** of 4 faults in this case

Experiments on friction compensation

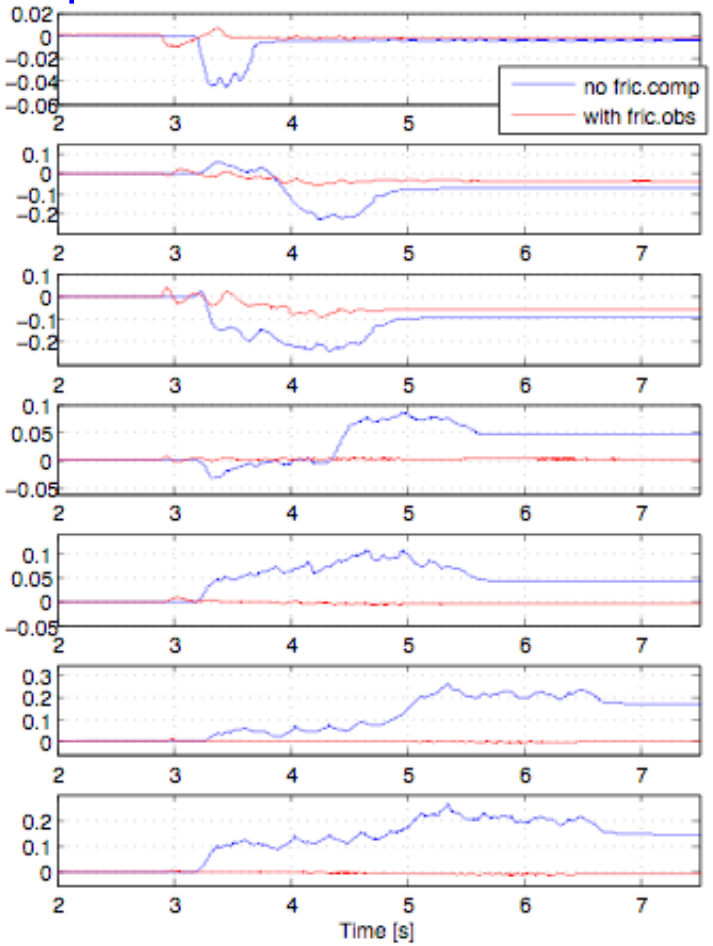
- results on the DLR 7R medical robot

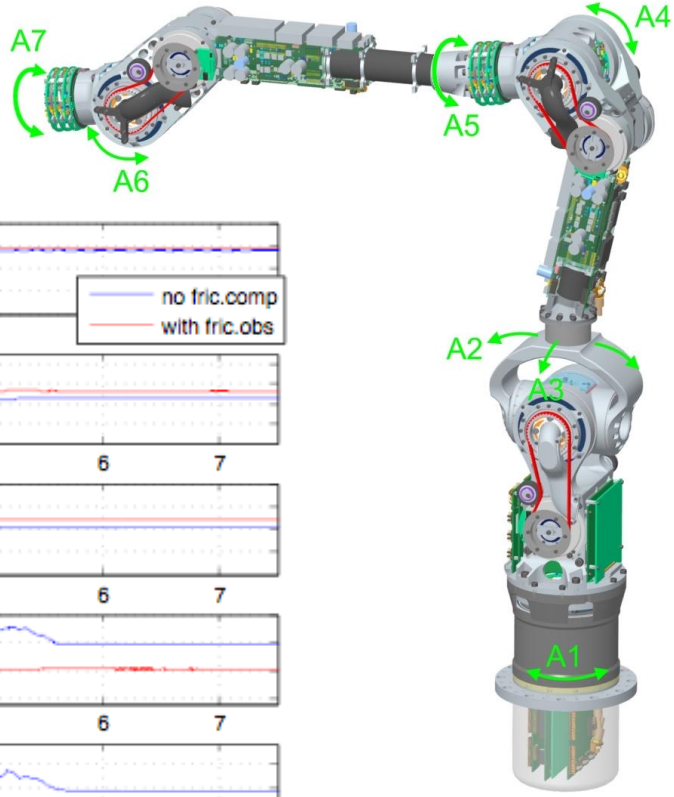
used then online in control law ...

friction estimate via residuals



position error





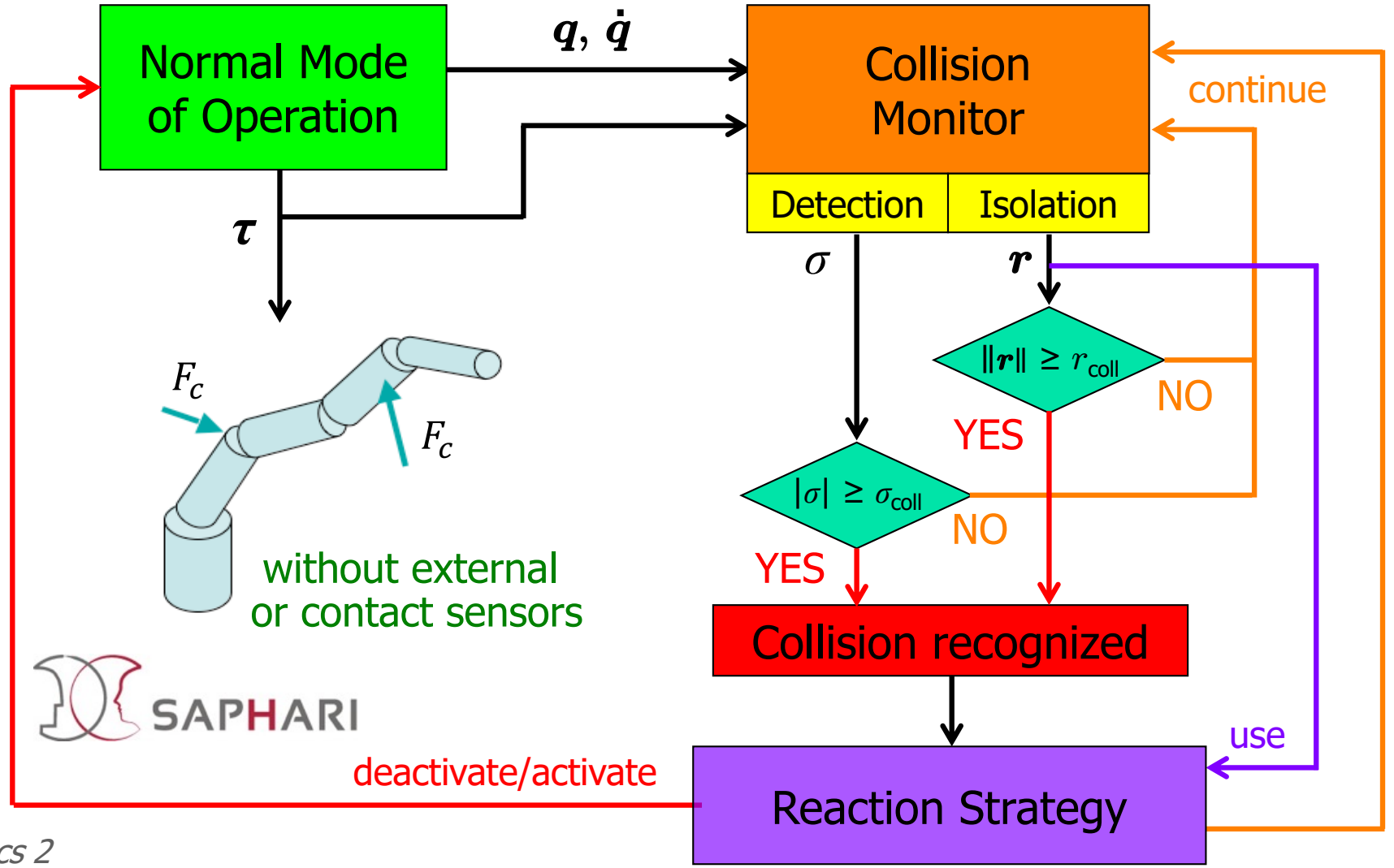
HD at the joints
⇒ elastic joint dynamic model

Collision detection and isolation

"sensorless" residuals

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau + \tau_c \qquad \tau_c = J_c^T(q)F_c$$

(both terms unknown)



Block diagram of the residual generator

momentum-based collision detection and isolation

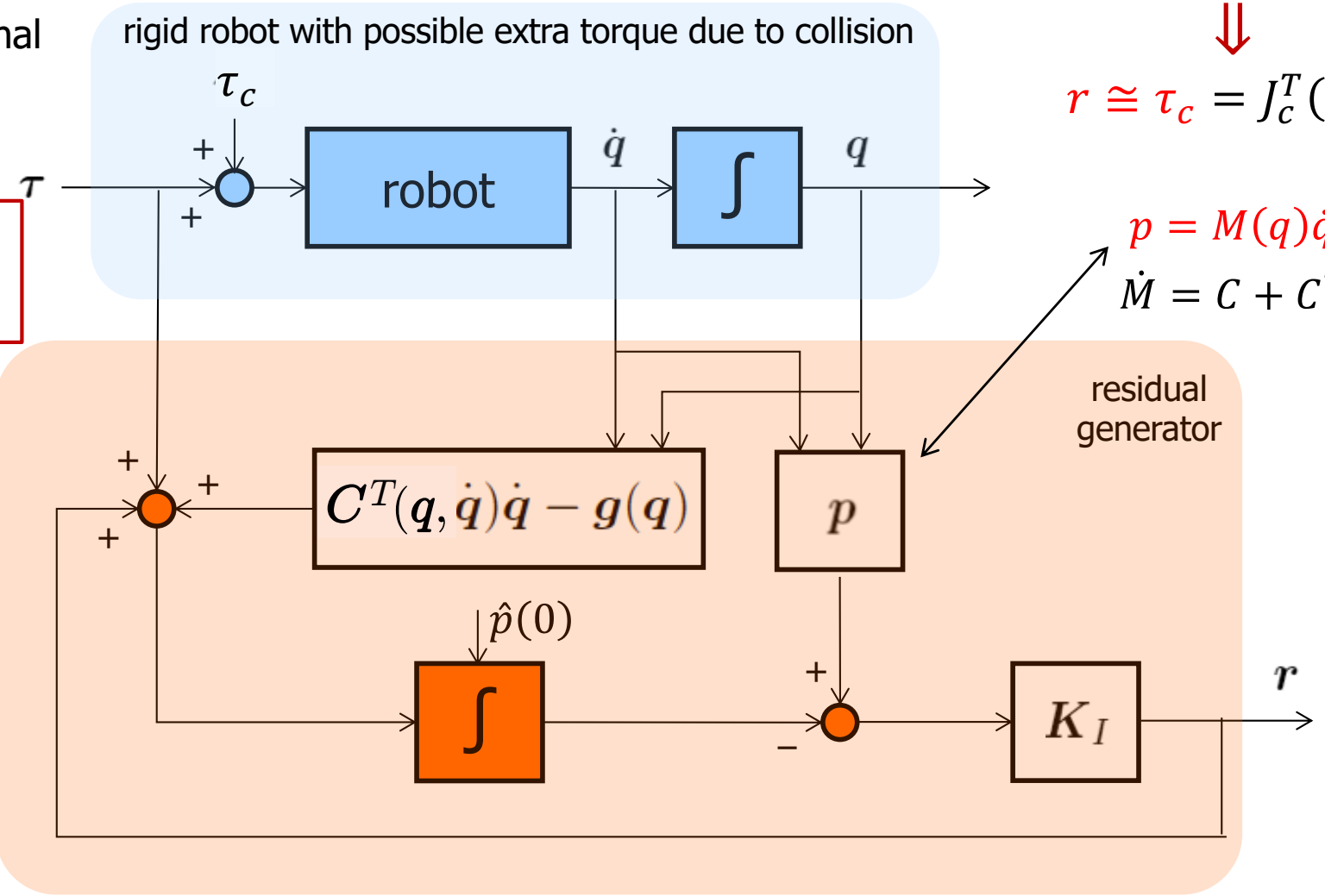
$$r(t) = K_I \left(p(t) - \int_0^t (\tau + C^T(q,\dot{q})\dot{q} + g(q) - r) ds - p(0) \right) \Rightarrow \boxed{\dot{r} = K_I(\tau_c - r)}$$

$$\Downarrow$$

$$r \cong \tau_c = J_c^T(q) F_c$$

$K_I > 0$, diagonal

more details in **pHRI** module on my web page!



initialization of integrators $\hat{p}(0) = p(0)$ (**zero** if robot starts at rest)

Collision detection and isolation

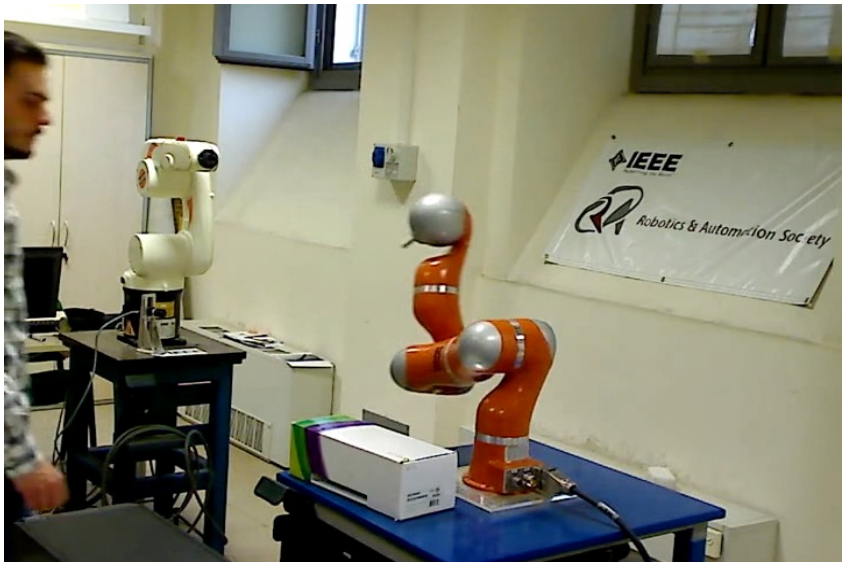
experiments with different robots over the years …

DLR III - **2006**

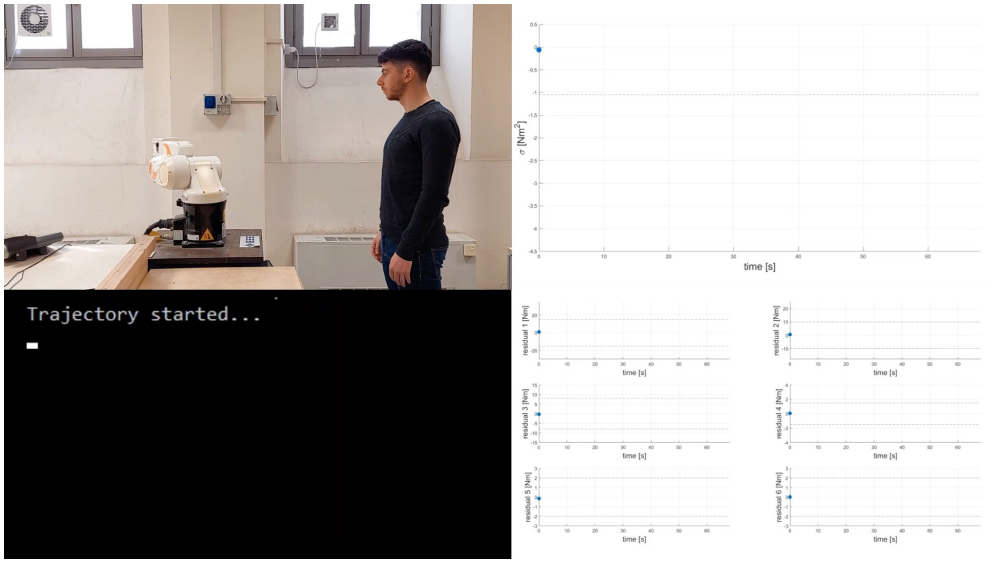


Neura - **2023**





LWR IV - **2015**



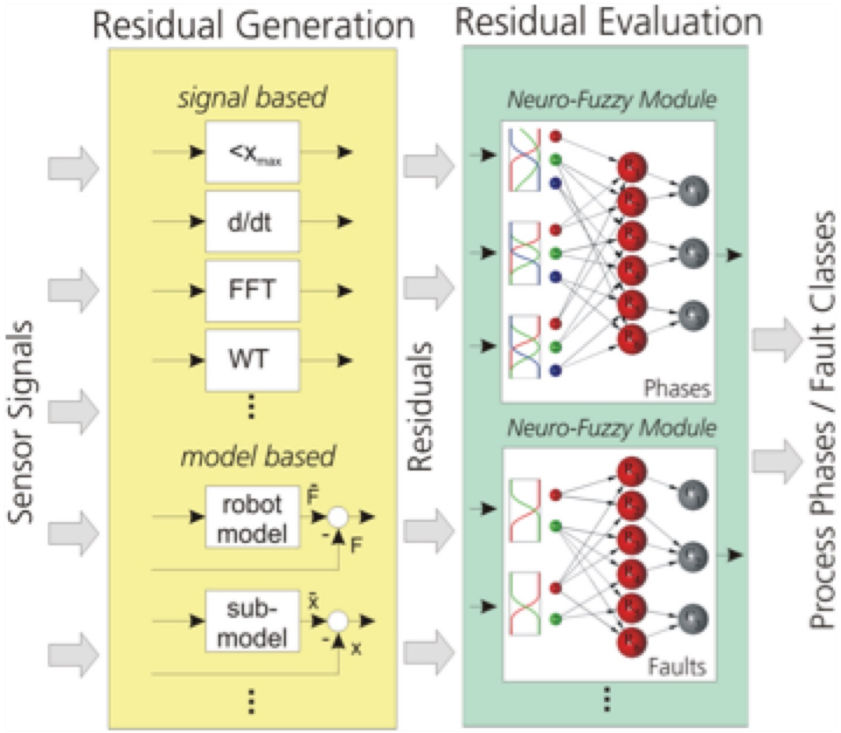
KUKA KR5 – **2022**

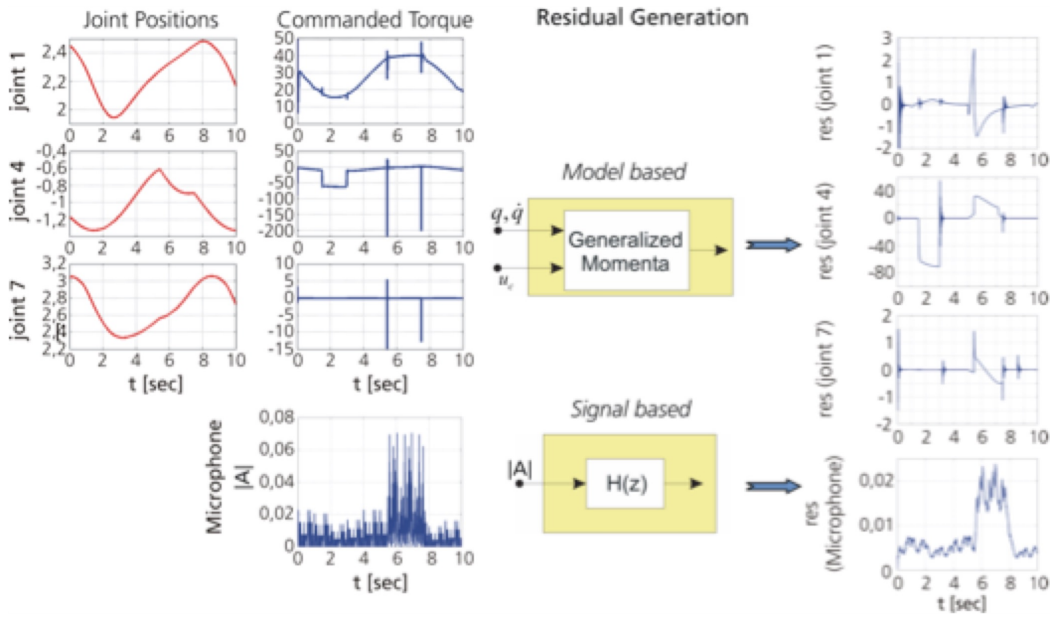
4 videos

Model- and signal-based FDI

- detection and isolation features can be enhanced by combining multiple sensor inputs and different approaches
  - model-based (exact, but require accurate models)
  - signal-based (approximate, but without special requirements)

so as to obtain the "best of both worlds"





SAPHARI

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